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CHE 321; Spring - 2020; Quiz - 1

100 Points

1. (25 points) What is the meaning of 'order of reaction'?

25 The order of the reaction is the sum of the powers of the concentrations of each species that participate in the reaction.
 in the rate equation. Tells you how many moles of reactant actually react in the reaction affect rate.
 example:
 $\text{rate} = k[A]^{\alpha}[B]^{\beta}$
 $\text{order} = \alpha + \beta$

2. (25 points) For a single liquid-phase reaction, which reactor will lead to a higher conversion: A plug-flow reactor (PFR) or a continuous stirred tank reactor (CSTRs)?

25 PFR

CSTR

3. (25 points) A catalytic CSTR reactor of volume V (m^3) has W (Mass) Kgs of the catalyst. What is the relationship between k_A and k'_A for this reactor? Note, the bulk density of the catalyst is $\rho_B = W/V$.

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$$k_A = \rho_B \cdot k'_A$$

$k_A = \frac{\text{rate}}{\text{vol. / m}^3}$

4. (25 points) Qualitatively, draw a Levenspiel's plot for two CSTRs in series.

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